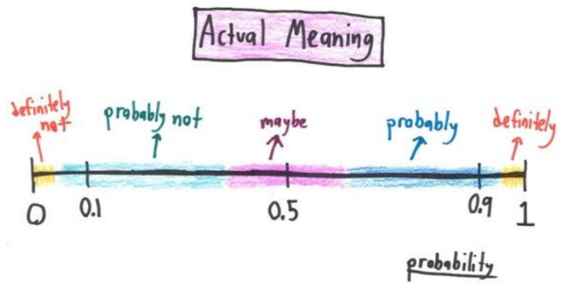
EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 7 - Lecture No 14 – February 25, 2026



Announcements!

1. Office hours (Ramadan) – Tuesday & Thursday 11:30 to 12:30 PM @ Room C-211 (or through meeting request)
2. Midterm Examination - March 3, 2026, from 03:25 PM to 04:55 PM at the Tariq Rafi Lecture Hall
   - ✓ 5 days to go! Practice Problems uploaded & Midterm Exam for Fall 2025 uploaded
3. Assignment 2 – due March 8, 2026
4. TA's EHSAS Hours

EHSAS TUTORS' RAMADAN SCHEDULE - Spring 2026							
Course	**Tutor**	**Email**	**Monday**	**Tuesday**	**Wednesday**	**Thursday**	**Friday**
Intro to Probability & Statistics	Rayyan Ali Shah	rs08770@st.habib.edu.pk	**2:00-3:00**		**2:00-3:00**		**12:40-1:40**
	Hiba Aiman Ali	ha09269@st.habib.edu.pk	**12:45-2:30**		**12:45-2:30**		**11:00-12:00**
	Zain Naqi	zn09224@st.habib.edu.pk		**9:00 - 10:30**	**11:45 - 12:45**	**9:00 - 10:30**	

Agenda for today

1. Quiz No 04 (on DRV)
2. Unit 4 – Discrete Random Variables

Unit 4
Discrete Random Variables

TAs -

EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics, Spring 2026

Meet Your TAs

Zain Naqi

About Me

I am a Computer Science Junior. My career interests include machine learning, deep learning, and reinforcement learning. Apart from academics, I do competitive programming and run a small e-commerce business. Mostly, you will find me in the library discussion area.

Ehsaas Hours

- Tue 9:30 - 11:30
- Thu 9:30 - 11:30

Contact Me

zno9224@st.habib.edu.pk or “Zain Naqi” on Teams

Meet Your TAs

Rayyan Shah

About Me

I am a Computer Science Junior. I am interested in bioinformatics, AI, and game development. Outside of academics, I also enjoy cooking and baking, and I always have some new recipe to recommend. You can usually find me in the library or tapal.

Ehsaas Hours

- Mon 4:00-5:00
- Wed 4:00-5:00
- Fri 3:30-4:30

Contact Me

rso8770@st.habib.edu.pk

Or “Rayyan Ali Shah” on Teams

Meet Your TAs

Hiba Aiman Ali

About Me

I am a Computer Engineering Junior with a minor in South Asian Music. My interests include bioinformatics, IoT, music and sustainability. I also sing as a part of the Habib University Orchestra. When I’m not in class, I’m most likely in the music room, singing or *trying to* learn an instrument.

Ehsaas Hours

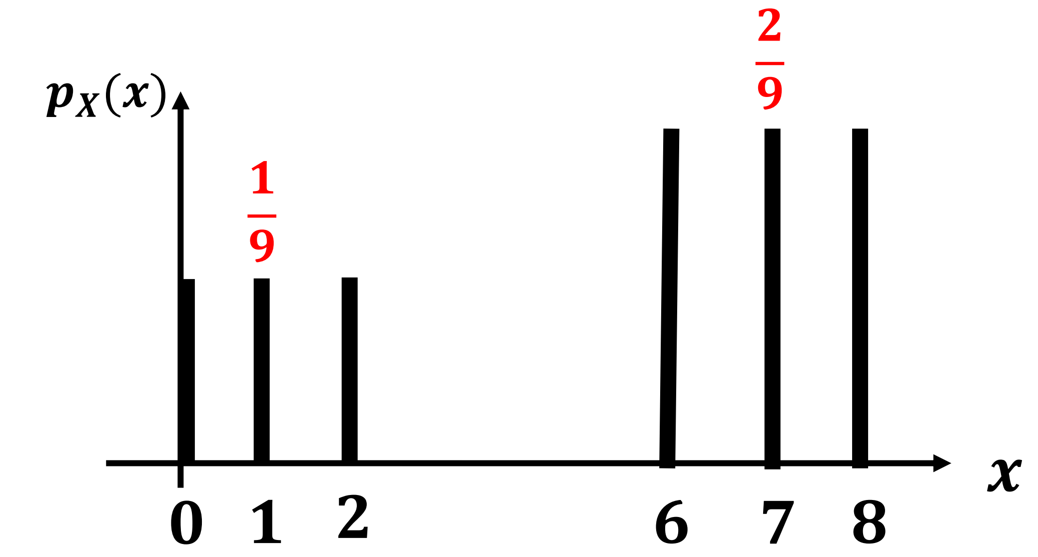
- Mon 2:30 - 4:00
- Wed 2:30 - 4:00
- Fri 12:00-1:00

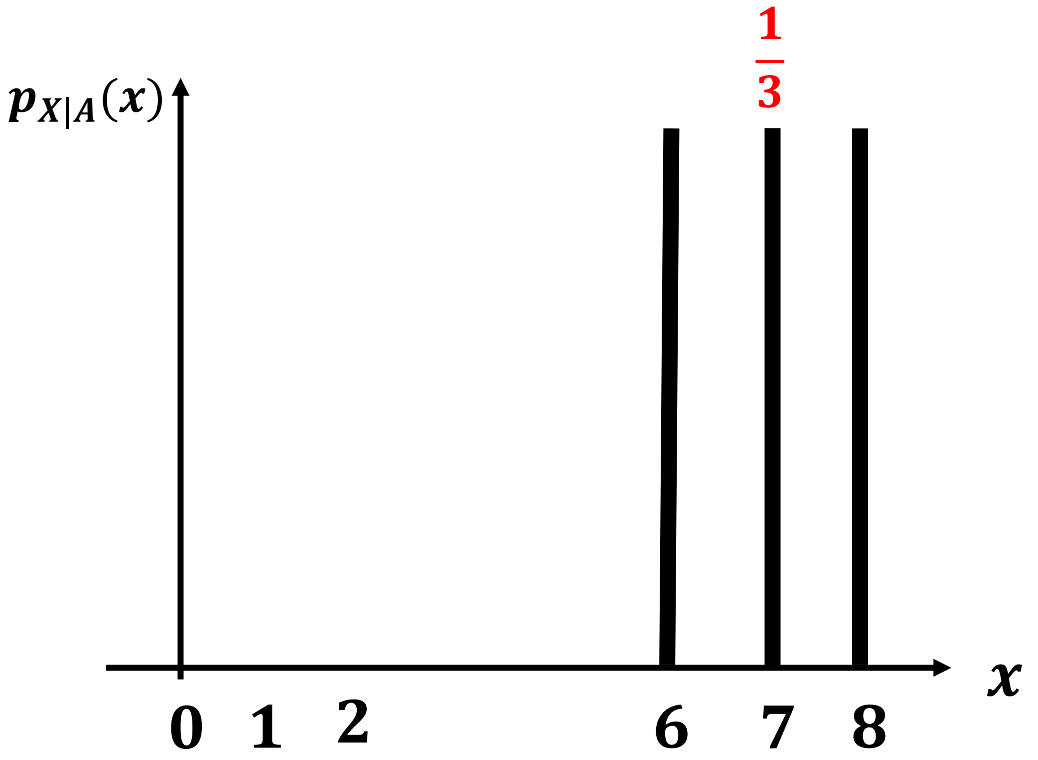
Contact Me

hao9269@st.habib.edu.pk on Teams or Outlook

$A = \{x > 2\}$

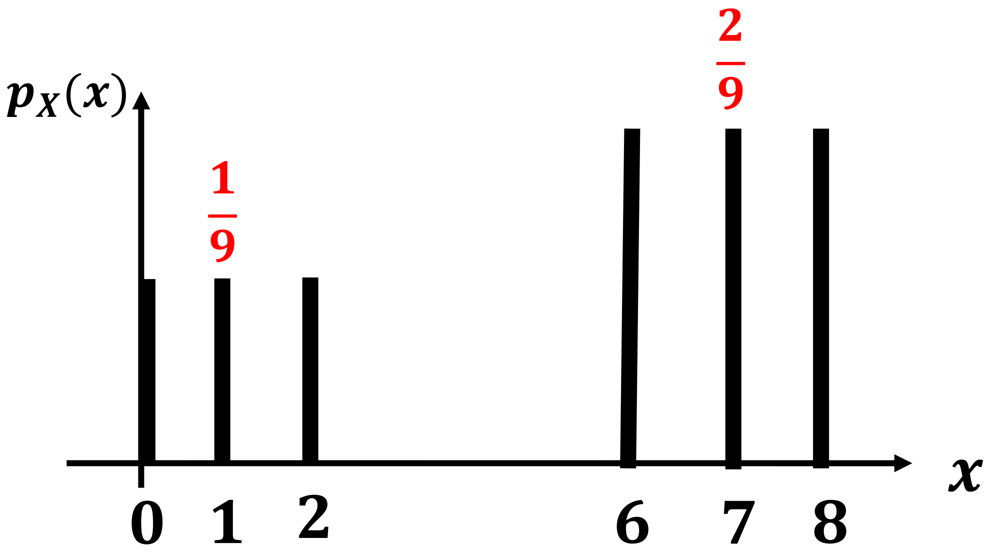
$P_{X|A}(x)$?

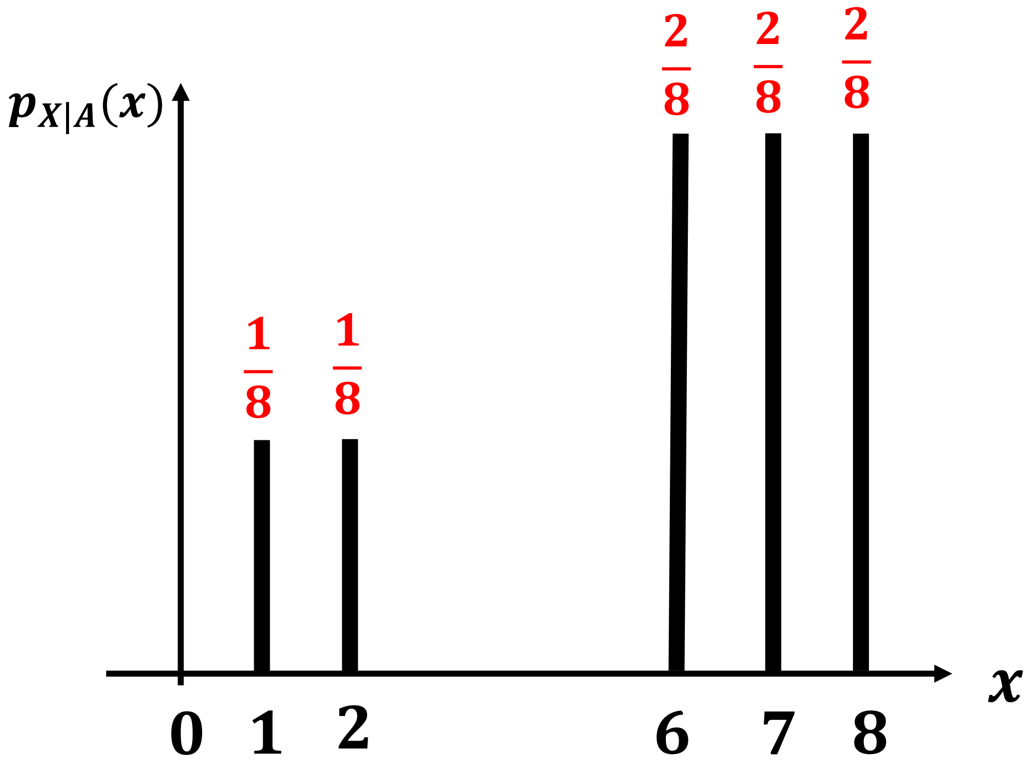


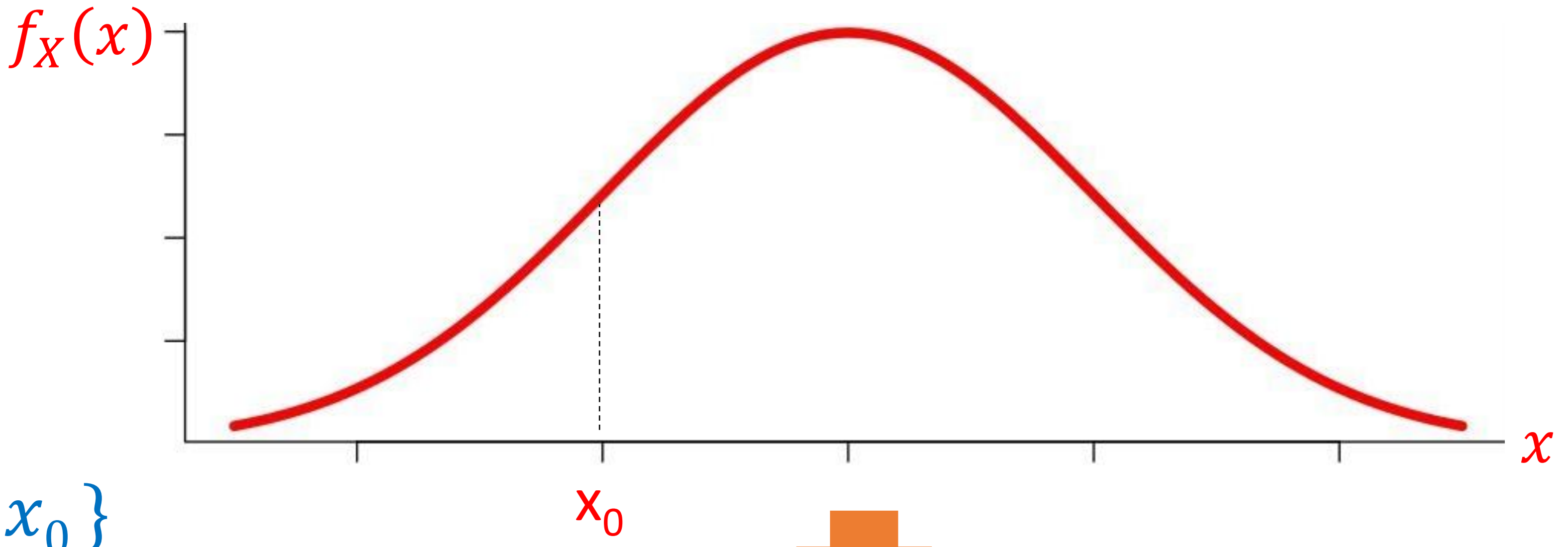


$A = \{\, x > 0\}$

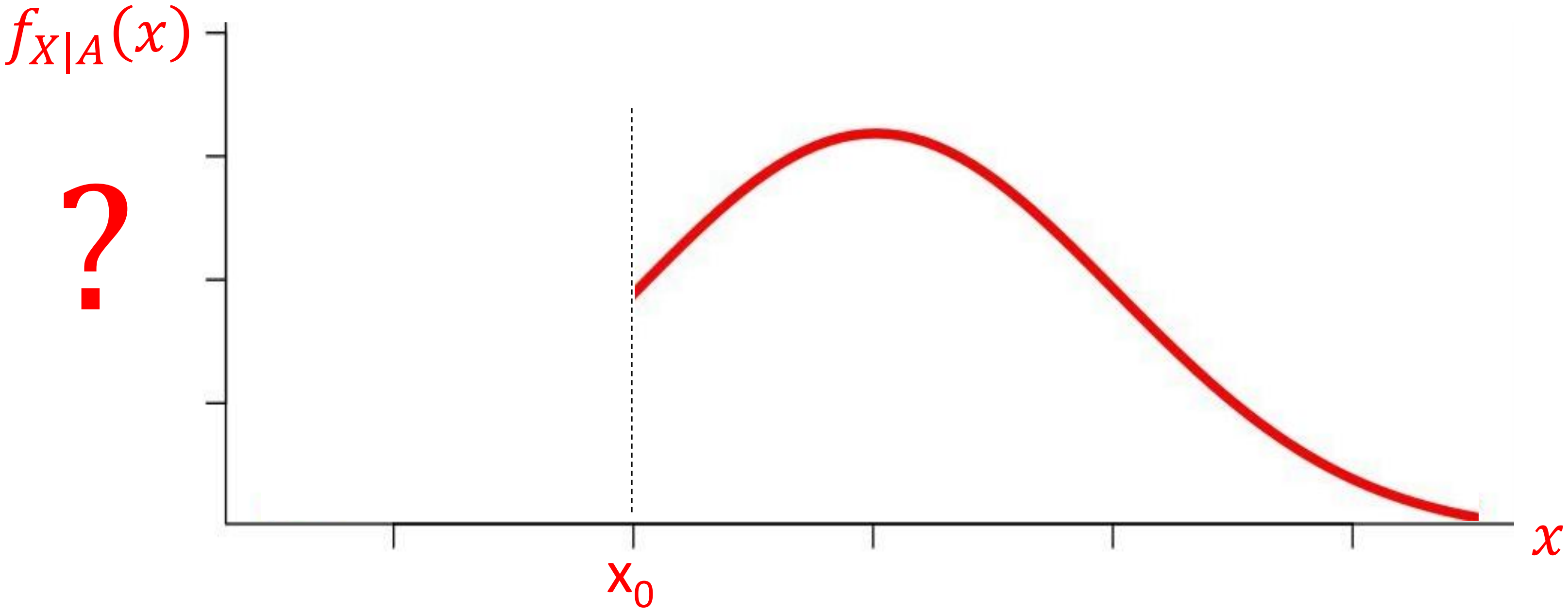
$P_{X|A}(x)$?

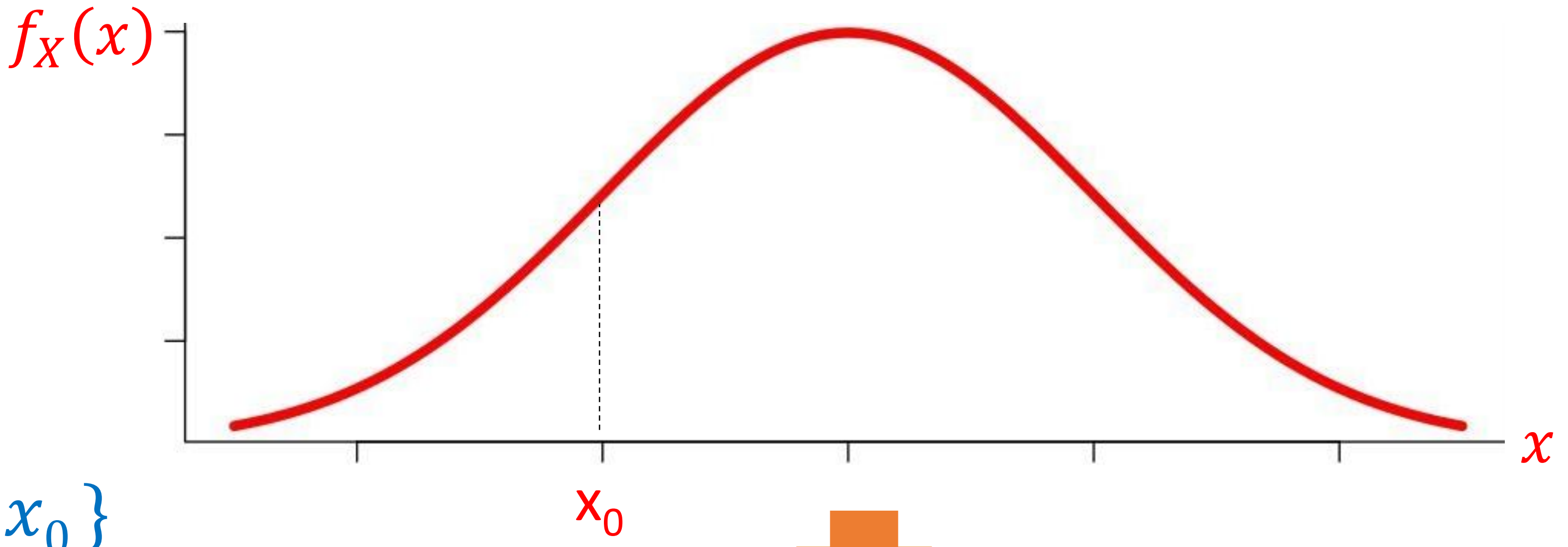

$p_X(x)$
$\frac{1}{9}$
$\frac{2}{9}$
x
0
1
2
6
7
8


$p_{X|A}(x)$
$\frac{2}{8}$
$\frac{2}{8}$
$\frac{2}{8}$
$\frac{1}{8}$
$\frac{1}{8}$
x
0
1
2
6
7
8



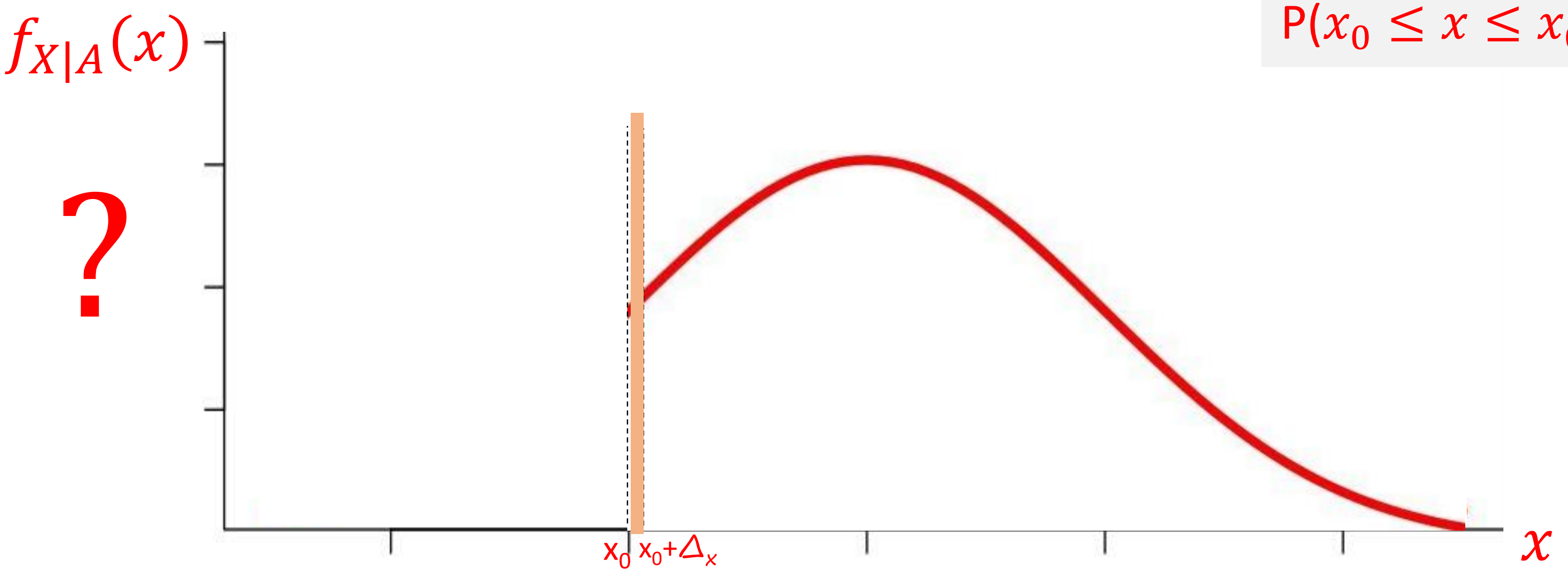
$A = \{ x > x_0 \}$

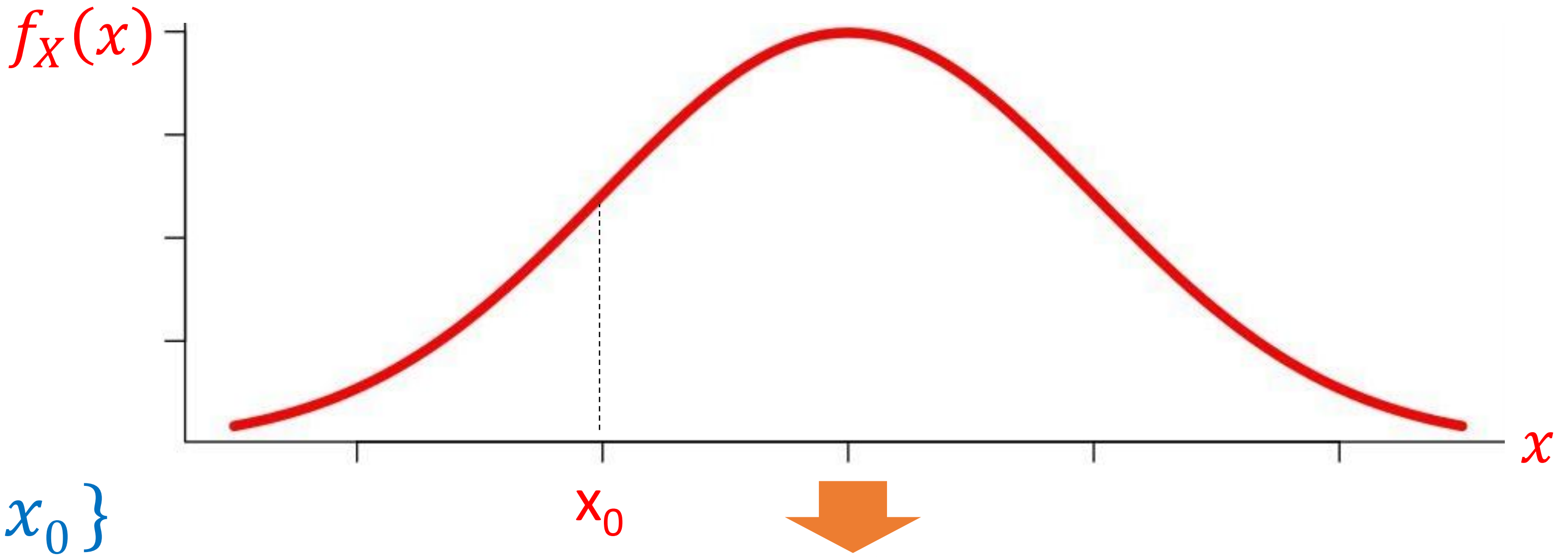




$$A = \{\, x > x_0 \,\}$$

$$\mathrm{P}(x_0 \leq x \leq x_0 + \Delta_x) \approx f_{X|\mathrm{A}}(x_0).\Delta_x$$





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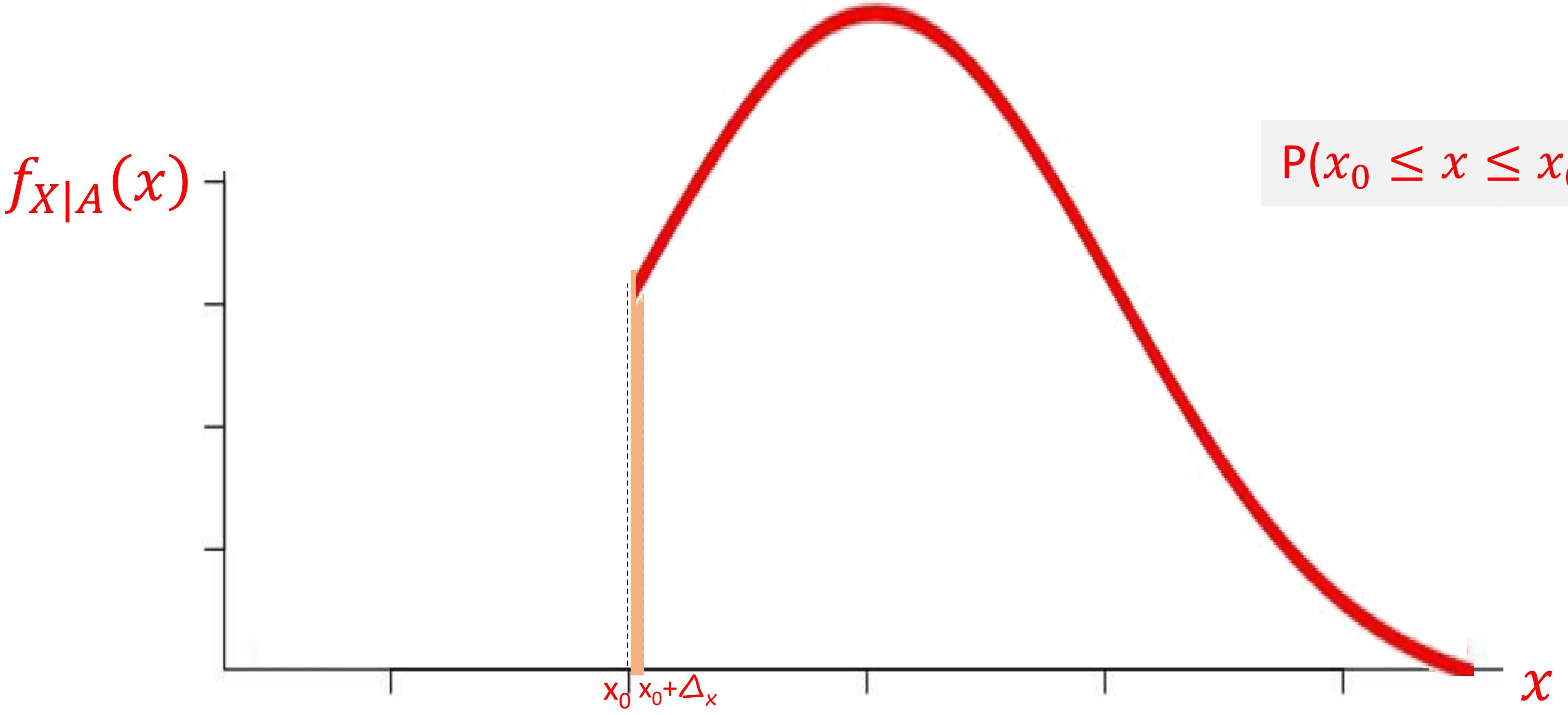
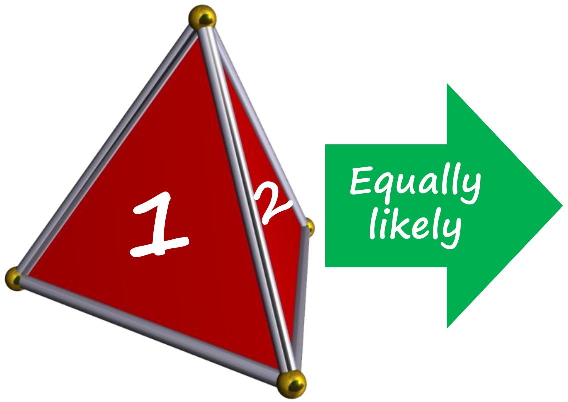


$$P(x_0 \le x \le x_0 + \Delta_x) \;\approx f_{X|\mathrm{A}}(x_0).\Delta_x$$

Exercise



Problem 3 (25 points): A Random Variable Z_n is defined as $Z_n = max\,(X_1, X_2, ..., X_n)$, where X_i's are outcomes of identical and independently distrubted tetrahedral die (4-sided). As an example $Z_2 = max\,(X_1, X_2)$ - meaning Z_2 is the maximum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

Part a (5 points): Calculate probability P($Z_{10} < 4$).

Part b (5 points): Calculate probability P($Z_3 = 2$).

Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2.

Part d (10 points): Find the Expected value, E $[Z_2|A]$, where event A = $\{X_1 = 3\}$.

Part a

$Z_{10} < 4 \Rightarrow max(X_1, \dots X_{10}) < 4)$

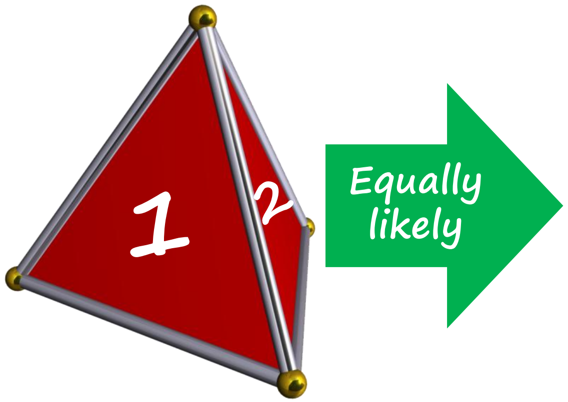
means zero No' of 4 in 10 throws

Probability of No 4 in One throw $= 1 - 1/4$

$= 3/4$

So $P(Z_{10} < 4) = \left(3/4\right)^{10} = 0.0563$

Exercise



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Part a (5 points): Calculate probability $\mathrm{P}(Z_{10} < 4)$.

Part b (5 points): Calculate probability $\mathrm{P}(Z_3 = 2)$.

Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2.

Part d (10 points): Find the Expected value, $\mathrm{E}\,[Z_2|A]$, where event A $= \{X_1 = 3\}$.

Part b

$$P(Z_3 = 2)$$

$$= \max(X_1, X_2, X_3) = 2$$

$\Downarrow$

Each Seq Probability $= \frac{1}{4^3}$

for $Z_3 = 2$

outcomes

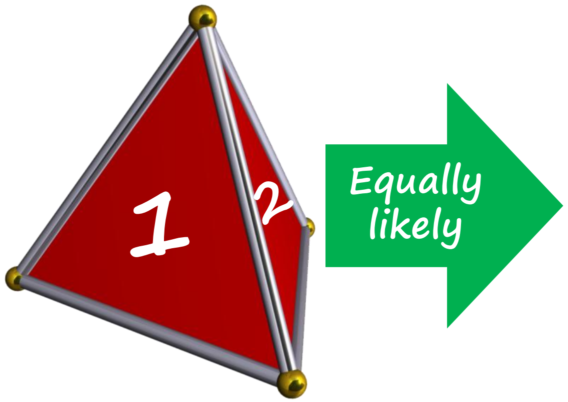
1, 1, 2
1, 2, 1
2, 1, 1 ✓
2, 2, 1 ✓
2, 1, 2 ✓
1, 2, 2
2, 2, 2 ✓

1	2	2
2	1	2
2	2	1

$$= 7 \times \frac{1}{4^3}$$

$$= 0.109$$

Exercise



Problem 3 (25 points): A Random Variable Z_n is defined as $Z_n = max\,(X_1, X_2, ..., X_n)$, where X_i's are outcomes of identical and independently distrubted tetrahedral die (4-sided). As an example $Z_2 = max\,(X_1, X_2)$ - meaning Z_2 is the maximum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

Part a (5 points): Calculate probability $\mathrm{P}(Z_{10} < 4)$.

Part b (5 points): Calculate probability $\mathrm{P}(Z_3 = 2)$.

Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2.

Part d (10 points): Find the Expected value, $\mathrm{E}\,[Z_2|A]$, where event $\mathrm{A} = \{X_1 = 3\}$.

Part c

$Z_2 = 1 \Rightarrow$

X_1	X_2
1	1

$1 \times \frac{1}{4^2}$

$Z_2 = 2 \Rightarrow$

1	2
2	1
2	2

$3 \times 1/4^2$

$Z_2 = 3 \Rightarrow$

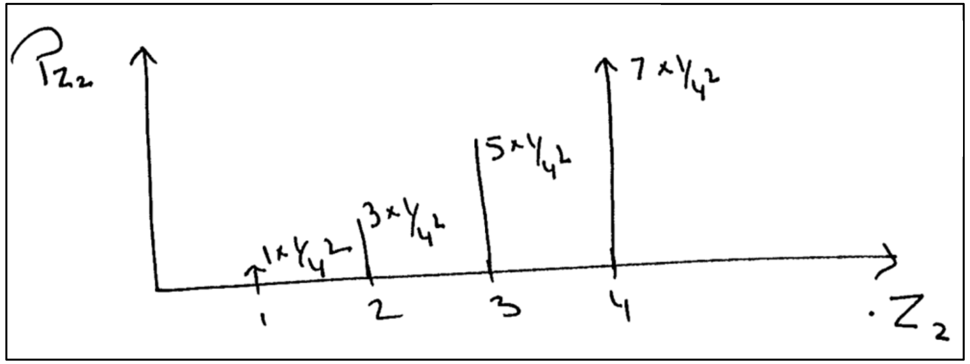
1	3
2	3
3	3
3	1
3	2

$5 \times 1/4^2$

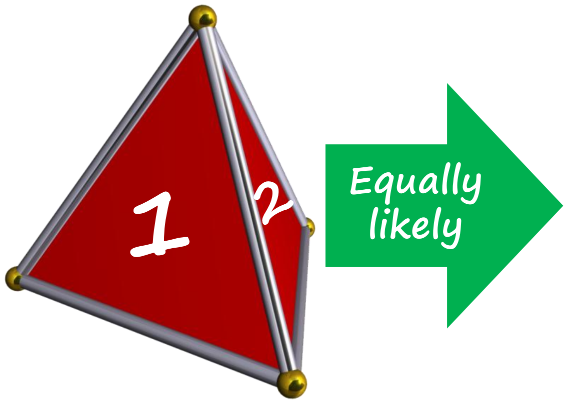
$Z_2 = 4 \Rightarrow$

1	4
2	4
3	4
4	4
4	1
4	2
4	3

$7 \times 1/4^2$



Exercise



Problem 3 (25 points): A Random Variable Z_n is defined as $Z_n = max\,(X_1, X_2, ..., X_n)$, where X_i's are outcomes of identical and independently distrubted tetrahedral die (4-sided). As an example $Z_2 = max\,(X_1, X_2)$ - meaning Z_2 is the maximum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

Part a (5 points): Calculate probability $P(Z_{10} < 4)$.

Part b (5 points): Calculate probability $P(Z_3 = 2)$.

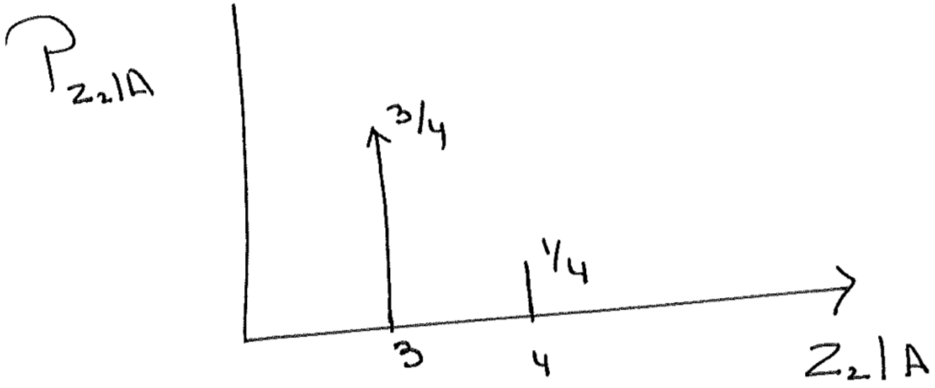
Part c (5 points): Calculate and plot the probability mass function, PMF of Z_2.

Part d (10 points): Find the Expected value, $E\,[Z_2|A]$, where event $A = \{X_1 = 3\}$.

Part d

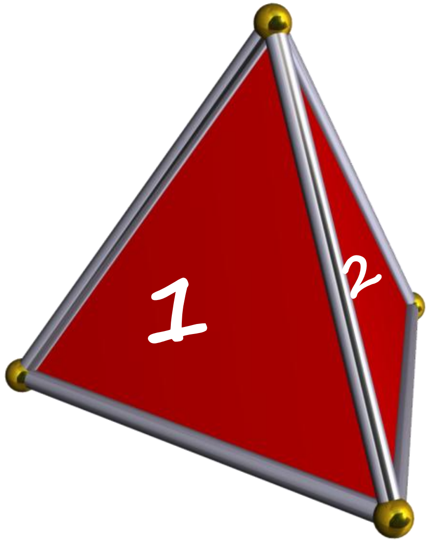
$$E\left[Z_2 \middle| (X_1 = 3)\right]$$

X_1	X_2	$P(X_2 = x)$	Z_2
3	1	1/4	3
3	2	1/4	3
3	3	1/4	3
3	4	1/4	4



$$E[Z_2|A] = 3 \times 3/4 + 4 \times 1/4$$

$$= \frac{9}{4} + \frac{4}{4} = 13/4$$



Exercise!

Equally likely

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \ldots; X_n)$, where X_i's are outcomes of identical and independently distributed tetrahedral die (4-sided & fair). As an example $Z_2 = \min(X_1; X_2)$ – meaning Z_2 is the minimum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

$$var(X) = E[X^2] - (E[X])^2$$

1. Calculate Variance of Z_1

2. Calculate Variance of Z_2

Quiz No 04